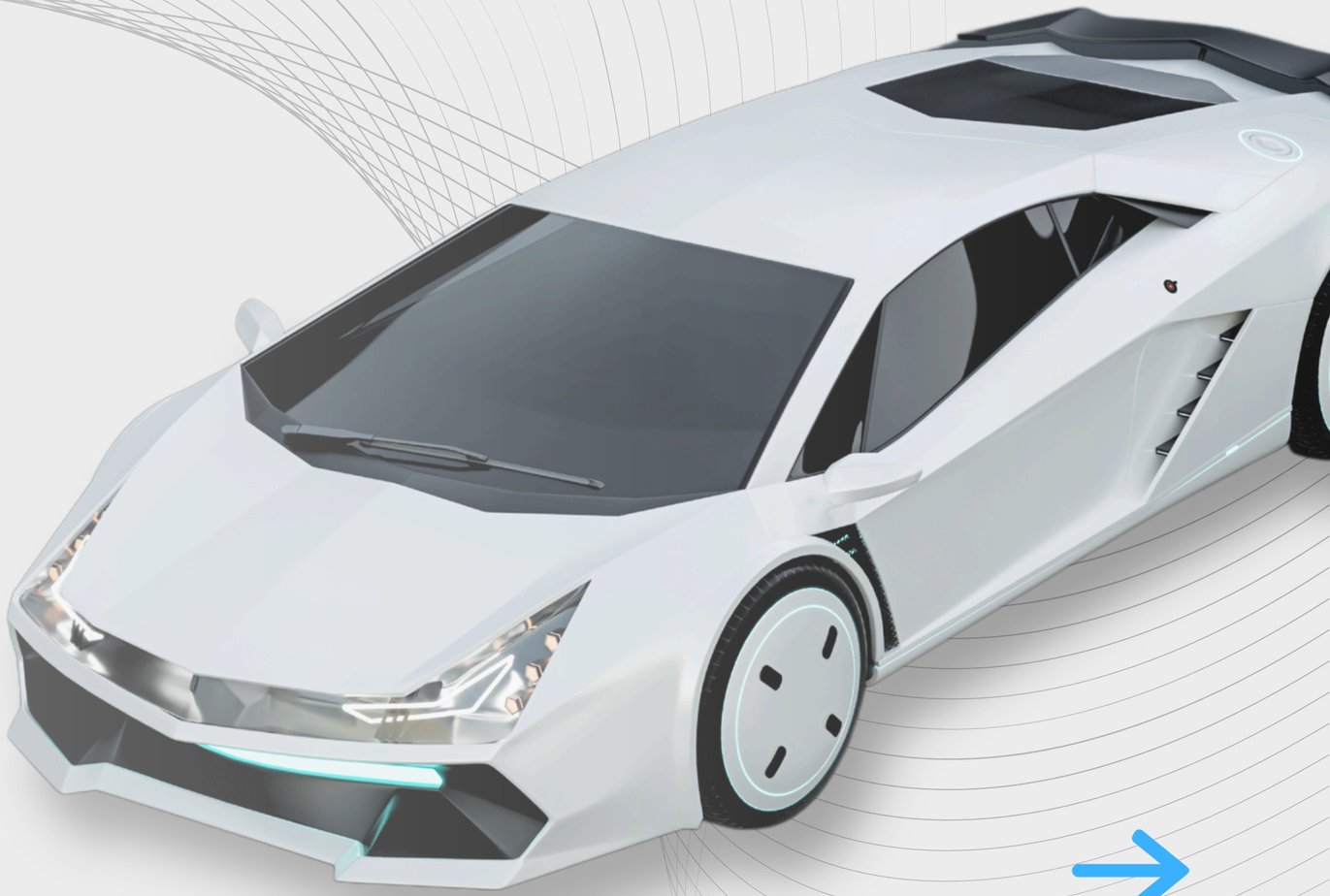
EV BHARAT

Sustainable, Smart, and Powerful



Presented by:

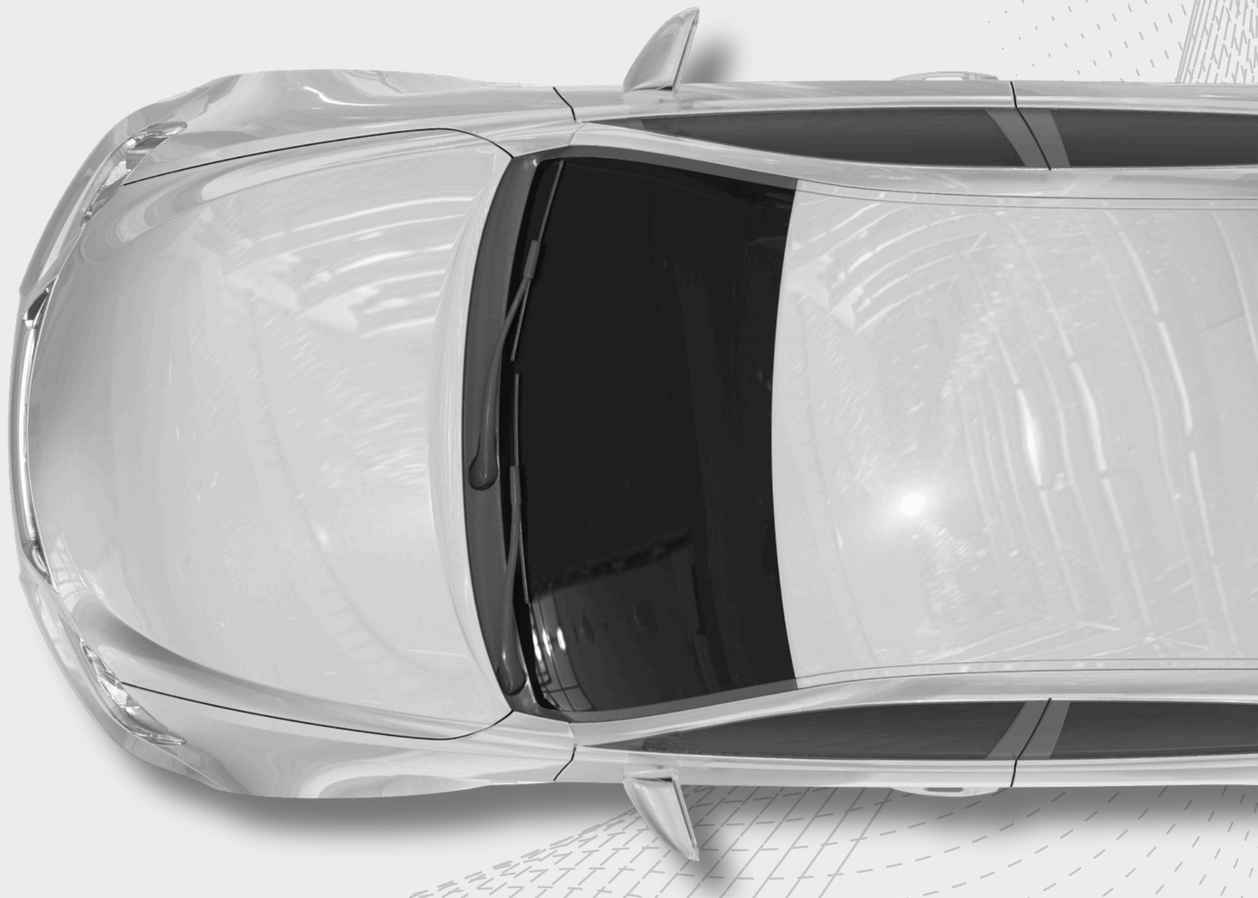
Karthik Janardhan

Made by:

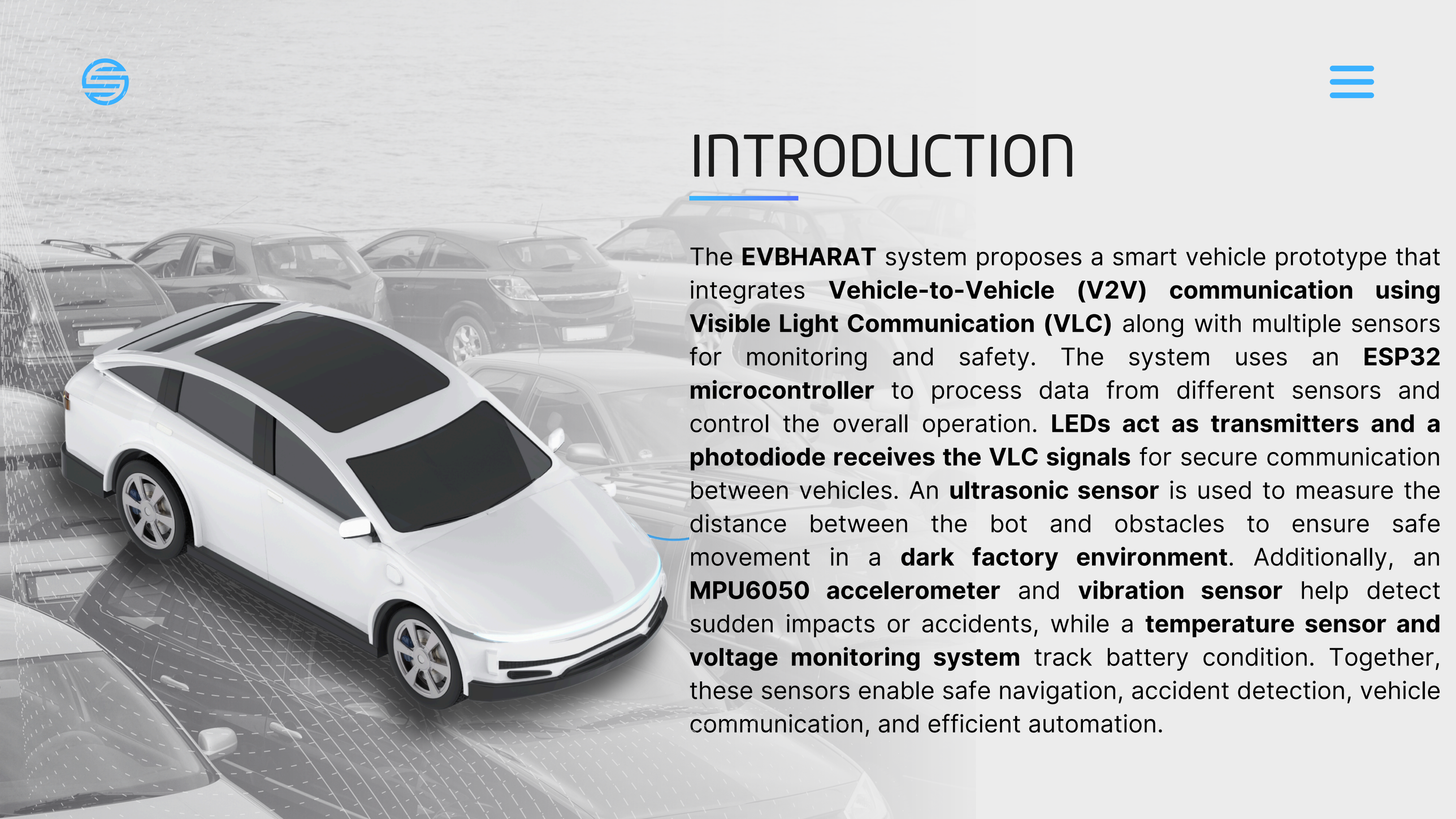
Anushka Kotal
Karthik Janardhan
Ashwin Vincent
Stash Lopes

PROBLEM STATEMENT

Existing vehicle communication systems mainly rely on RF technologies, which often face issues such as spectrum congestion, electromagnetic interference, and security risks. Additionally, many systems do not effectively integrate accident detection and vehicle monitoring within a single platform. There is also a growing need for intelligent communication systems in **dark factory** environments, where automated robots and vehicles must operate and communicate efficiently with minimal human intervention.



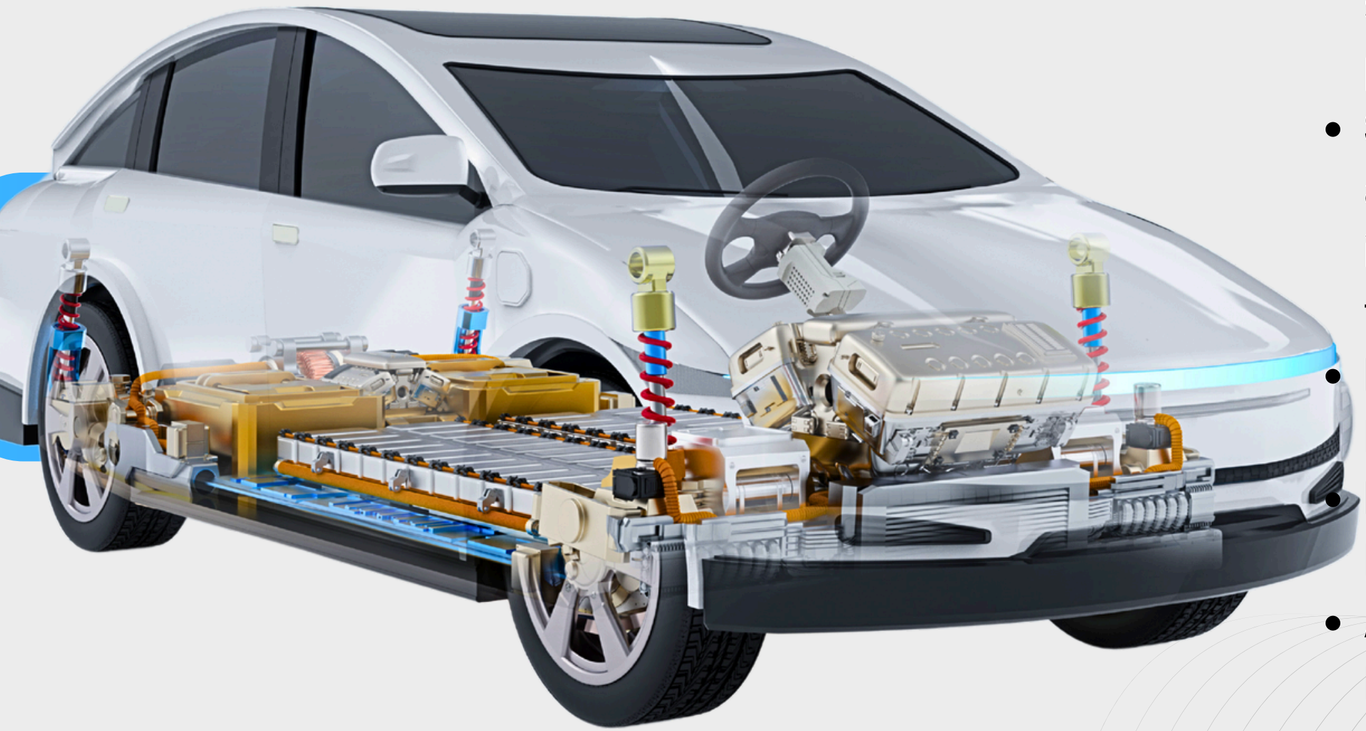
LETS GET STARTED



INTRODUCTION

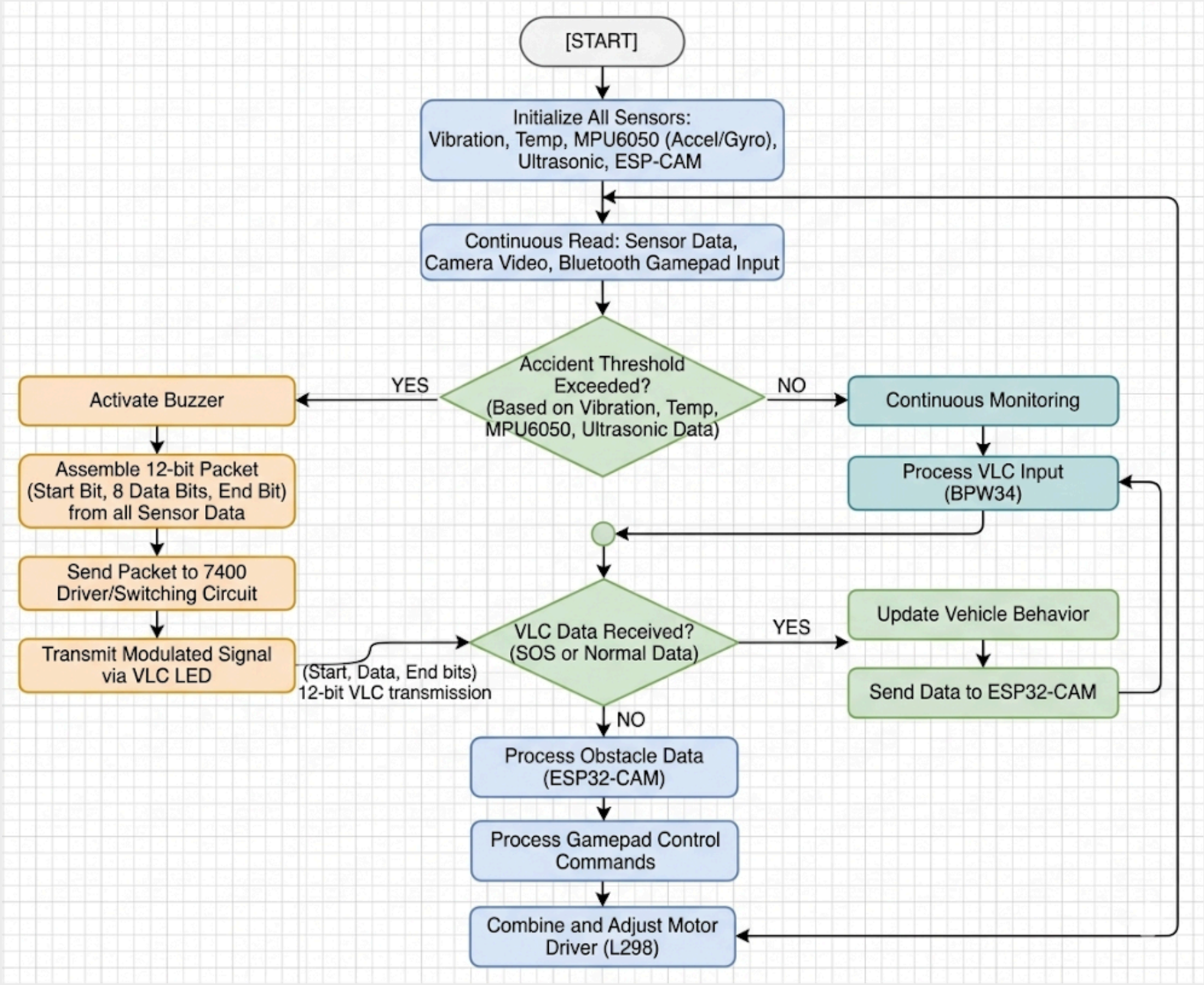
The **EVBHARAT** system proposes a smart vehicle prototype that integrates **Vehicle-to-Vehicle (V2V) communication using Visible Light Communication (VLC)** along with multiple sensors for monitoring and safety. The system uses an **ESP32 microcontroller** to process data from different sensors and control the overall operation. **LEDs act as transmitters and a photodiode receives the VLC signals** for secure communication between vehicles. An **ultrasonic sensor** is used to measure the distance between the bot and obstacles to ensure safe movement in a **dark factory environment**. Additionally, an **MPU6050 accelerometer** and **vibration sensor** help detect sudden impacts or accidents, while a **temperature sensor and voltage monitoring system** track battery condition. Together, these sensors enable safe navigation, accident detection, vehicle communication, and efficient automation.

OUR SOLUTION

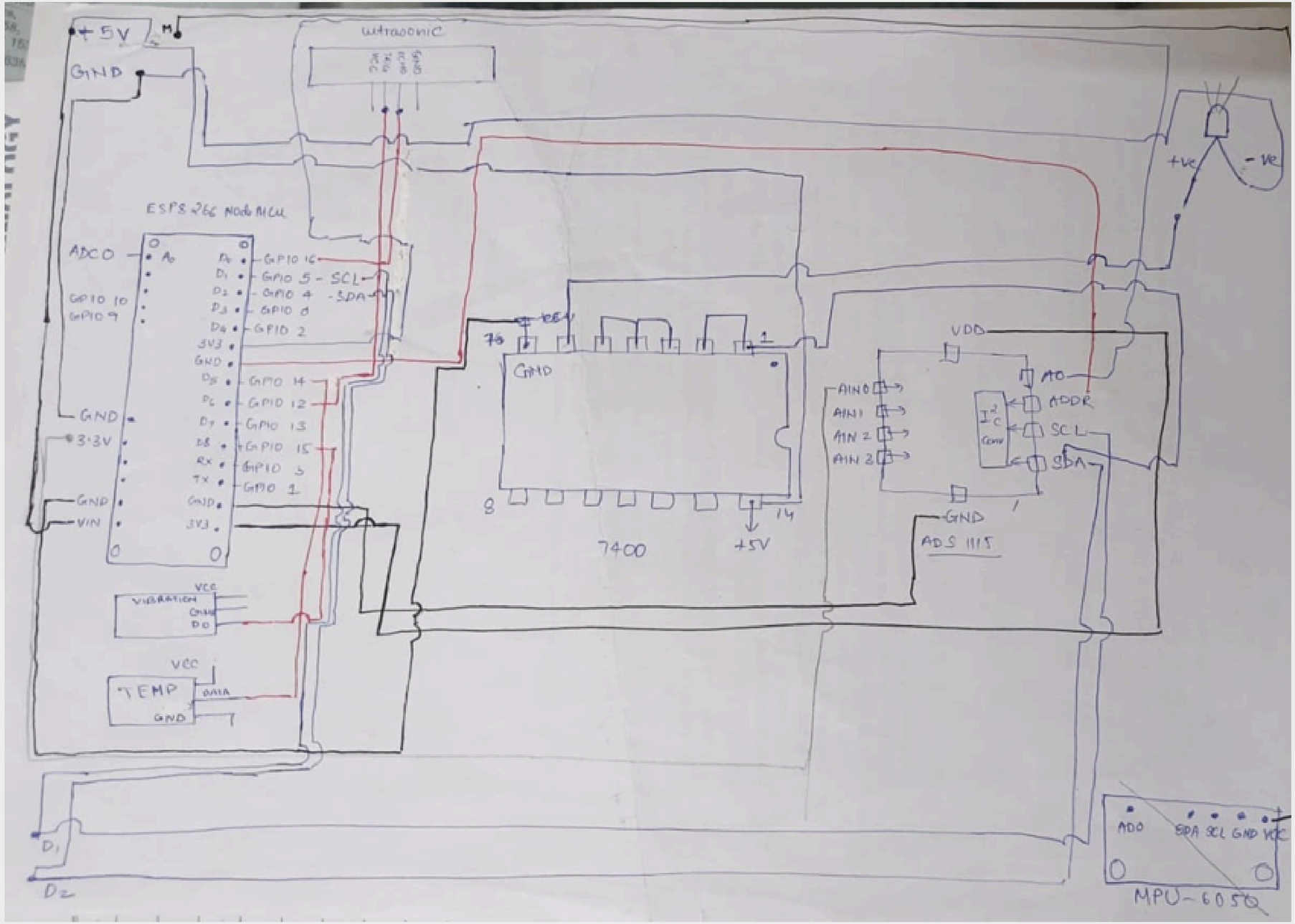


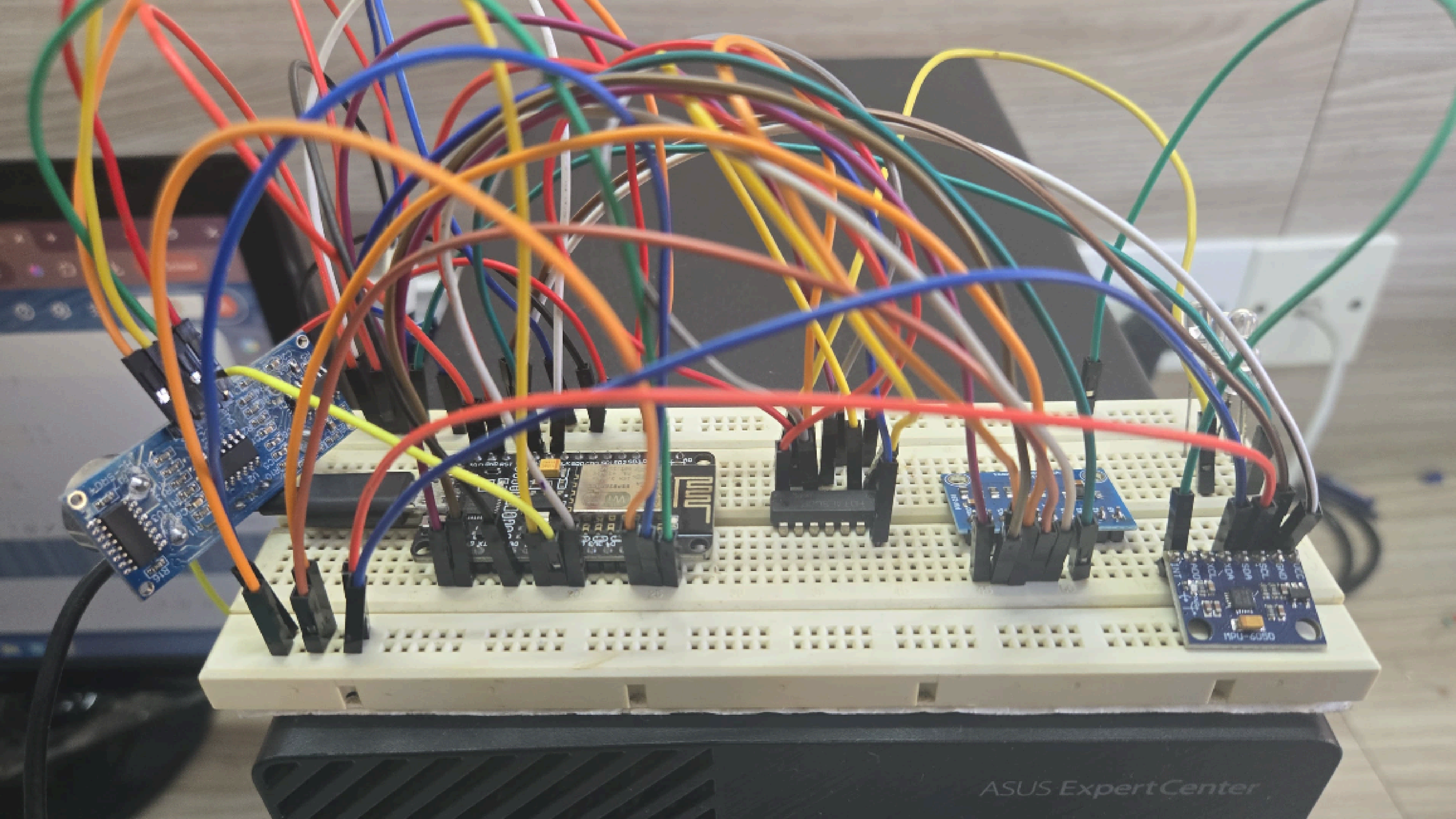
- **Microcontroller & Processing:** ESP32 DevKit V1 is used as the main controller for sensing, processing, and communication. The ESP32-CAM module enables camera-based object detection and monitoring.
- **Communication System:** VLC is implemented using a high-brightness LED as the transmitter and a BPW34 photodiode as the receiver. 7400 NAND gates perform signal modulation, and the ADS1115 ADC converts the photodiode analog signals to digital data.
- **Sensors:** MPU6050 accelerometers are used for motion and accident detection. A vibration sensor detects impacts, and a temperature sensor monitors battery temperature.
- **Mobility & Actuation:** A L298 motor driver controls the DC motors responsible for robotic vehicle movement.
- **Power Supply:** The system is powered by three 3.7V Li-ion batteries.
- **Alert Mechanism:** A buzzer provides audio alerts, while LEDs provide visual warnings during system events.

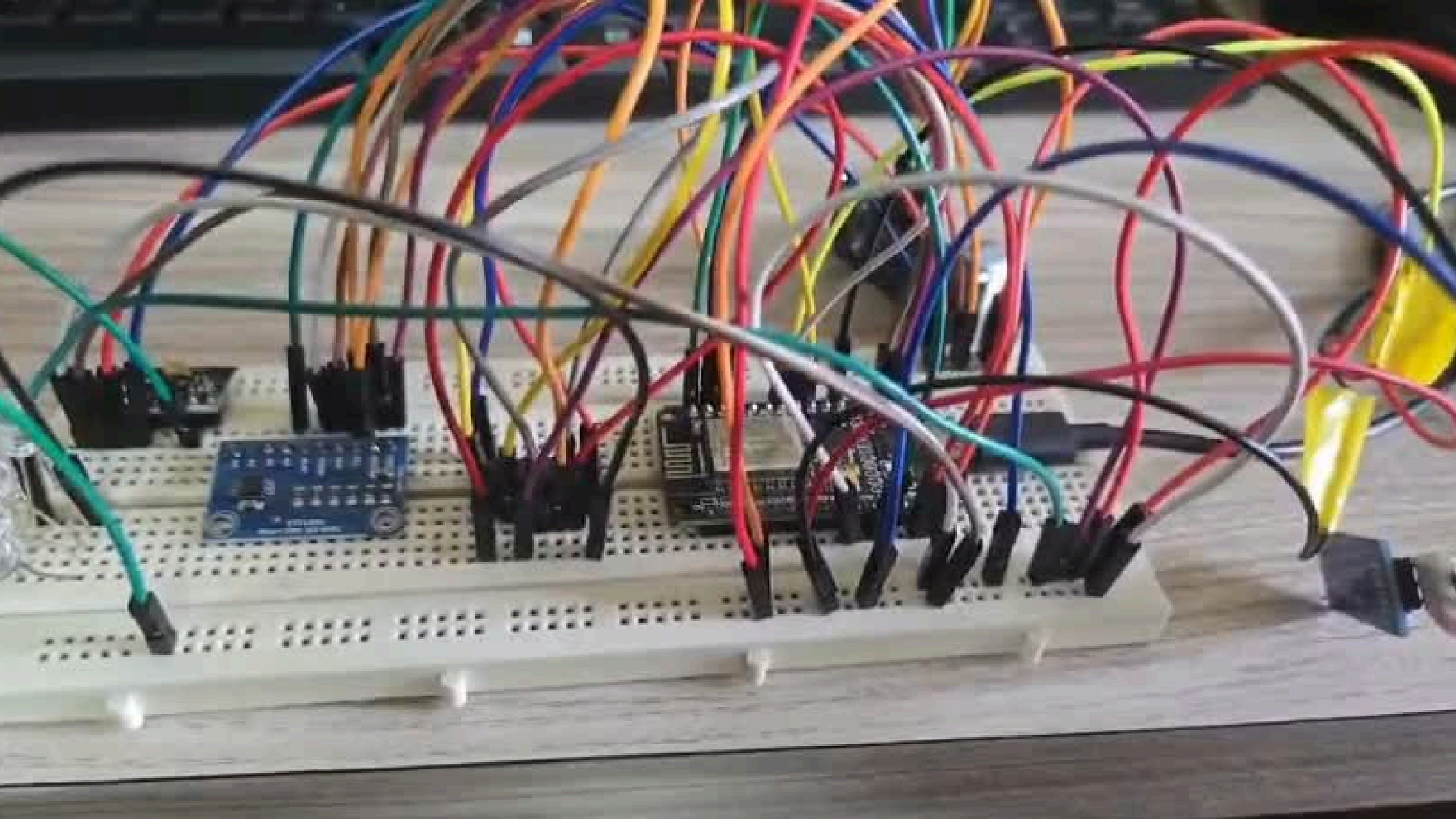
FLOWCHART


[START]
Initialize All Sensors: Vibration, Temp, MPU6050 (Accel/Gyro), Ultrasonic, ESP-CAM
Continuous Read: Sensor Data, Camera Video, Bluetooth Gamepad Input
Accident Threshold Exceeded? (Based on Vibration, Temp, MPU6050, Ultrasonic Data)
YES
Activate Buzzer
Assemble 12-bit Packet (Start Bit, 8 Data Bits, End Bit) from all Sensor Data
Send Packet to 7400 Driver/Switching Circuit
Transmit Modulated Signal via VLC LED
(Start, Data, End bits) 12-bit VLC transmission
NO
Continuous Monitoring
Process VLC Input (BPW34)
VLC Data Received? (SOS or Normal Data)
YES
Update Vehicle Behavior
Send Data to ESP32-CAM
NO
Process Obstacle Data (ESP32-CAM)
Process Gamepad Control Commands
Combine and Adjust Motor Driver (L298)

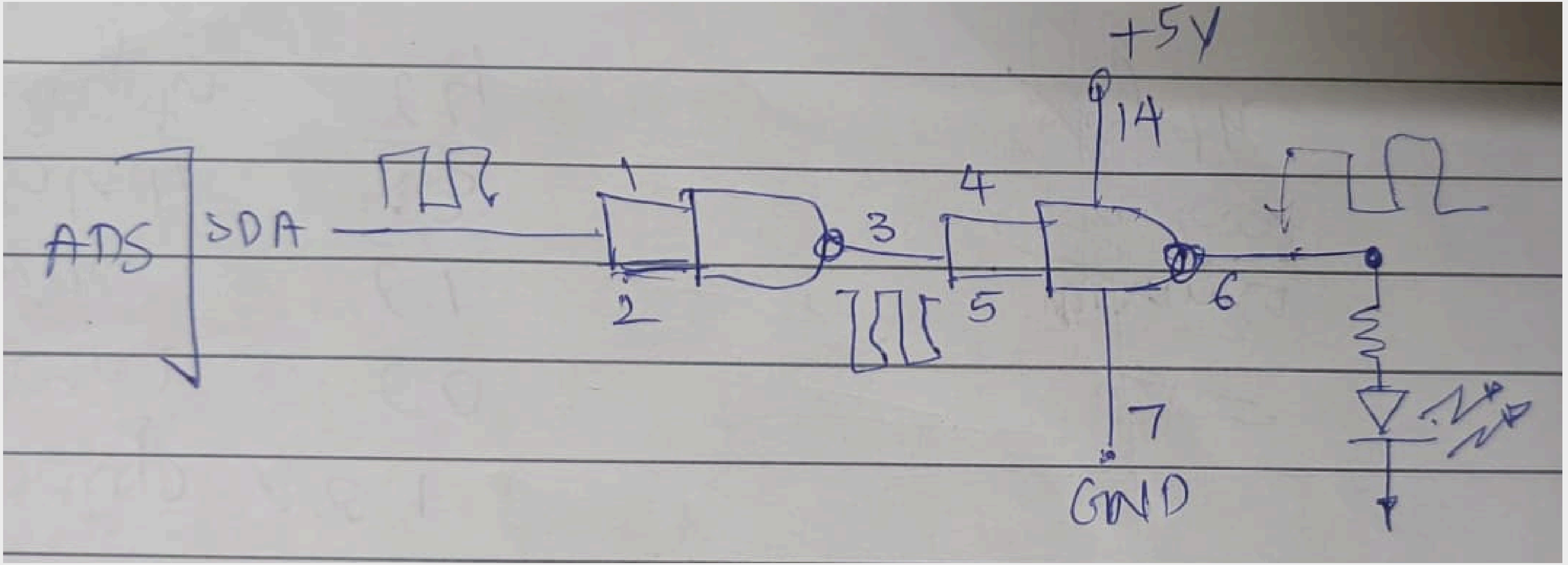
TRANSMITTER CIRCUIT DIAGRAM



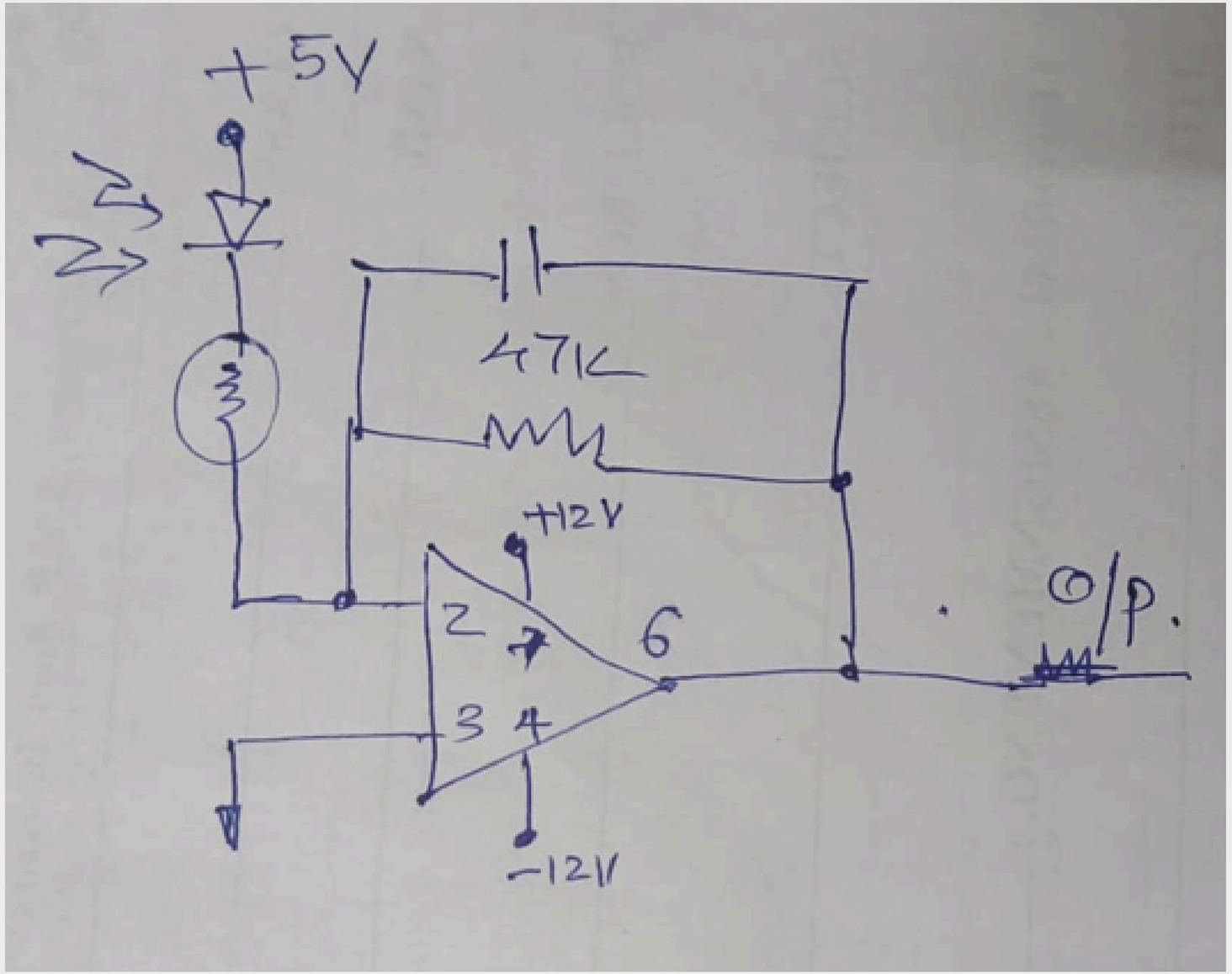

MPU-6050
ASUS ExpertCenter



TRANSMITTER 7400 IC CKT



RECEIVER CIRCUIT DIAGRAM



PROBLEMS FACED:

- WE DID THE CONNECTION OF THE RECEIVER CIRCUIT IE. SHOWN IN SLIDE NO. 10. BUT WE DIDN'T GET THE OUTPUT. IT GAVE GARBAGE VALUES OR 0 BITS CONTINUOUSLY.

- WE GOT 3 SOLUTIONS FROM MRINMOYEE MA'AM,

I) BY **INCREASING THE GAIN**, IE, WE REPLACED THE OLD RESISTOR WITH ANOTHER RESISTOR. BUT INCREASING THE GAIN ALSO DIDN'T WORK. THE POWER EMITTED IS VERY LITTLE (2.36V OR 2.40V). THE RECEIVER CANNOT CAPTURE IT. THE RECEIVER SIDES ESP MODULE WANTS POWER (3.3V) TO DEBUG AND THE DATA RATE IS LOW.

II) TO **INCREASE THE SENSITIVITY** OF BPW34 SO THAT IT CAPTURES THE DATA AT A FASTER RATE.

III) **RF AND CF.**

IV) TO CHANGE PHOTODIODE WITH BPW34:
- BPW32 PHOTODIODE CAN BE USED
- SFH303FA
- SOLAR PANELS

V) TRIAL ERROR METHOD: **REDESIGNING THE CIRCUIT** IE. TO CHANGE THE WHOLE RECEIVER CIRCUIT IE. TO CHANGE THE PHOTODIODE

PROBLEMS FACED:

FROM JOSEPH SIR:

SINCE IT IS SENSOR DATA, YOU DON'T NEED HIGH DATA RATE. SO USE A LARGER AREA PHOTO DETECTOR. YOU COULD USE RED LEDS OR ARRAYS.

OR

YOU COULD USE HIGH POWER (1 W OR 3W) WHITE LEDS WHICH WILL GIVE YOU MORE POWER.

- WE USED ALL SENSORS, SO THE POWER WAS LESS USING 1 LED. THERE WAS NOISE ALSO WHICH IT WAS CAPTURING SO WE REMOVED ALL SENSORS, KEPT ONLY 1 SENSOR AND 1 LED.

- WE SAW IE. LIGHT GLOWS BRIGHTER COMPARED TO EARLIER... WE CHECKED IN CRO WHERE THE SIGNAL STRENGTH WENT FROM 0.5 V TO 2.3V, SO POWER INCREASED AND NOISE CONSTANT.

THANK YOU
FOR YOUR TIME AND VALUABLE INSIGHTS

